Modeling of Rotational Mechanical Systems
(转动机械系统的建模)

Main Contents of Course (课程主要内容)

Mechanical (机械)

Electrical (电路)

Electro-Mechanical (机–电)

Fluid (流体)

Thermal (热系统)

Dynamic Modeling (动态建模)

Model structure (模型结构)

System Identification(系统辨识)

Parameter estimation (参数估计)

Control System Design(控制系统设计)

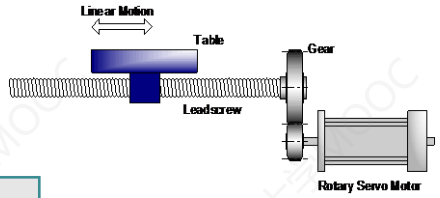
System Analysis (系统分析)

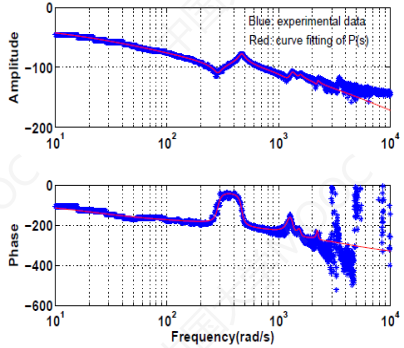
Root-locus design (根轨迹设计)

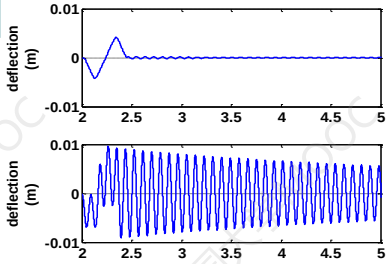
Pole placement design (极点配置)

Design limitations (性能局限)

Architectural issues (控制结构)





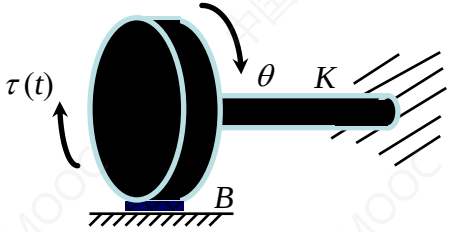


Variables

- θ: angular displacement (角位移)[rad]
- ω: angular velocity (角速度)[rad/sec]
- α: angular acceleration (角加速度)[rad/sec^2]

$$\frac{d}{dt}\theta = \dot{\theta} = \omega$$

$$\dot{\omega} = \frac{d}{dt}\left(\frac{d}{dt}\theta\right) = \frac{d^2}{dt^2}\theta = \ddot{\theta} = \alpha$$



- τ: torque (力矩) [Nm]
- p: power (功率) [Nm/sec]
- w: work (功) (energy (能量)) [Nm]

  1 [Nm] = 1 [J] (Joule)

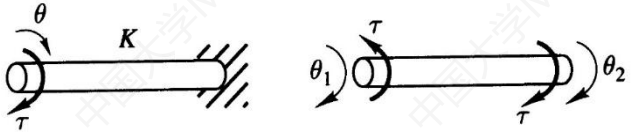
$$p = \tau \cdot \omega = \tau \cdot \dot{\theta} = \frac{d}{dt} w$$

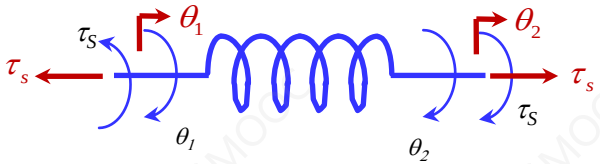
$$w(t_1) = w(t_0) + \int_{t_0}^{t_1} p(t)\, dt$$

$$= w(t_0) + \int_{t_0}^{t_1} (\tau \cdot \dot{\theta})\, dt$$

Basic Rotational Modeling Elements

- **Spring (弹簧)**
  - Stiffness Element (刚度元件)



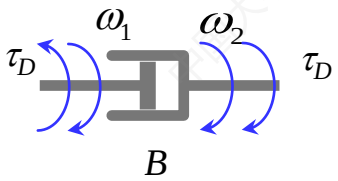


$$\tau_S = K(\theta_2 - \theta_1)$$

  - Analogous to Translational Spring (类似于平动弹簧)
  - Stores Potential Energy (储存势能)

    e.g., shafts (如弹性轴)

- **Damper (阻尼器)**
  - Friction Element (摩擦元件)



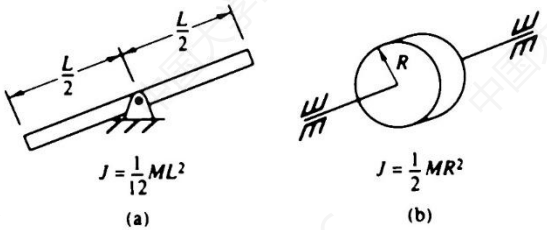
$$\tau_D = B(\dot{\theta}_2 - \dot{\theta}_1) = B(\omega_2 - \omega_1)$$

  - Analogous to Translational Damper (类似于平动阻尼器)
  - Dissipate Energy (耗散能量)

    e.g., bearings, bushings (如轴承，轴套), …

Basic Rotational Modeling Elements

- **Moment of Inertia (转动惯量)**
  - Inertia Element (惯性元件)



$$J\ddot{\theta} = \sum_i \tau_i$$

  - Analogous to Mass in Translational Motion (类似于平动运动的质量)
  - Stores Kinetic Energy (储存动能)

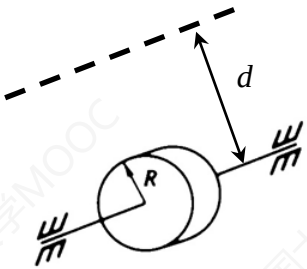
- **Parallel Axis Theorem (平行轴定理)**

$$J = J_O + Mr^2$$

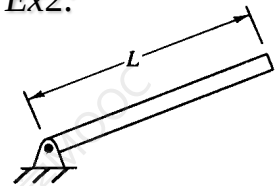
*Ex*1:

$$J = J_O + M\,r^2$$

$$= \frac{1}{2}MR^2 + Md^2$$



*Ex*2:



$$J = J_O + M\,r^2$$

$$= \frac{1}{12}ML^2 + M\left(\frac{L}{2}\right)^2$$

$$= \frac{1}{3}ML^2$$

Interconnection Laws

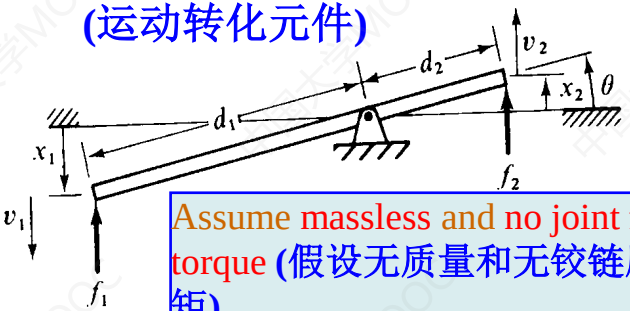
- **Newton's Second Law (牛顿第二定律)**

$$\frac{d}{dt}\underbrace{(J\omega)}_{\text{Angular Momentum}} = J\ddot{\theta} = \sum_i \tau_{EXTi}$$

- **Newton's Third Law (牛顿第三定律)**
  - Action & Reaction Torque (作用力&反作用力)

- **Lever (杠杆)**

Motion Transformer Element (运动转化元件)



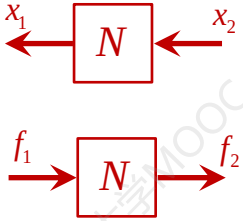
Assume massless and no joint friction torque (假设无质量和无铰链摩擦力矩)

$$\begin{aligned} x_1 &= d_1 \sin\theta \\ x_2 &= d_2 \sin\theta \end{aligned} \Rightarrow \frac{x_1}{x_2} = \frac{d_1}{d_2} = N$$

$$0 = J\ddot{\theta} = -f_1 d_1 \cos\theta + f_2 d_2 \cos\theta$$

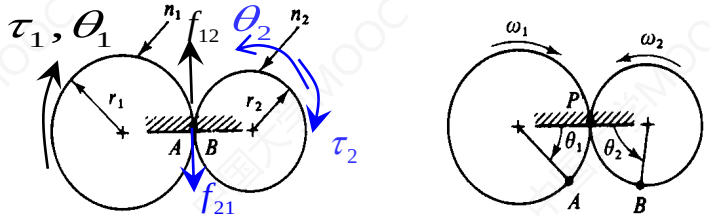
$$\Rightarrow f_1 d_1 \cos\theta = f_2 d_2 \cos\theta$$

$$\Rightarrow \frac{f_1}{f_2} = \frac{d_2}{d_1} = \frac{1}{N}$$



Motion Transformer Elements

- **Ideal Gears** (理想齿轮)



negligible moment of inertia and friction torques

Gear 1 $0 = J_1\ddot{\theta}_1 = \sum \tau = \tau_1 - f_{12}r_1$

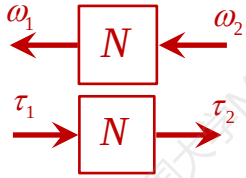
Gear 2 $0 = J_2\ddot{\theta}_2 = \sum \tau = -\tau_2 + f_{21}r_2$

$$\begin{cases} \tau_1 = f_{12}r_1 \\ \tau_2 = f_{21}r_2 \end{cases} \Rightarrow \frac{\tau_1}{\tau_2} = \frac{f_{12}r_1}{f_{21}r_2} = \frac{r_1}{r_2} = \frac{1}{N}$$

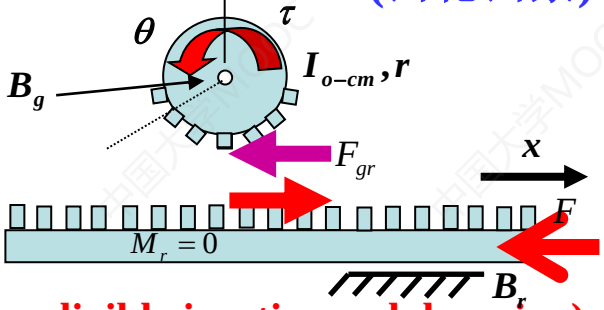
$$r_1\omega_1 = r_2\omega_2 \Rightarrow \frac{\omega_2}{\omega_1} = \frac{r_1}{r_2} = \frac{1}{N}$$

Energy Conservation
（能量转换）

$$\tau_1\omega_1 = \tau_2\omega_2$$



- **Rack and Pinion** (齿轮齿条)



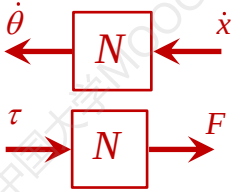
Ideal case (negligible inertias and damping)
理想情况(忽略惯量和阻尼)

$$0 = I_{o-cm}\ddot{\theta} = \tau - F_{gr}r$$

$$0 = M_r\ddot{x} = F_{gr} - F$$

$$\Rightarrow \frac{F}{\tau} = \frac{1}{r}$$

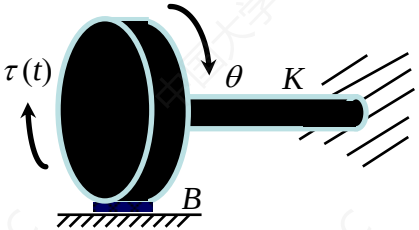
$$\dot{x} = r\dot{\theta} \Rightarrow \frac{\dot{\theta}}{\dot{x}} = \frac{1}{r}$$



$$\tau\dot{\theta} = F\dot{x}$$

Example

- **Derive a model (EOM) for the following system (推导模型)**

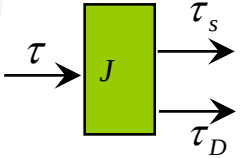


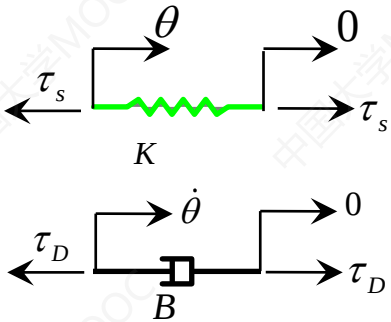
Translational

$\Rightarrow$

Equivalent

(等效平动模型)





$$J\ddot{\theta} = \sum \tau = \tau + \tau_s + \tau_D$$

$$\tau_s = K(0-\theta) = -K\theta$$

$$\tau_D = B(0-\dot{\theta}) = -B\dot{\theta}$$

$$J\ddot{\theta} = \tau - K\theta - B\dot{\theta}$$

$$J\ddot{\theta} + B\dot{\theta} + K\theta = \tau$$

Energy Distribution

- **EOM of a Mass-Spring-Damper System (质量-弹簧-阻尼系统的运动方程)**

$$\underbrace{J\ddot{\theta}}_{\substack{\text{Contribution}\\\text{of Inertia}}} + \underbrace{B\dot{\theta}}_{\substack{\text{Contribution}\\\text{of the Damper}}} + \underbrace{K\theta}_{\substack{\text{Contribution}\\\text{of the Spring}}} = \underbrace{\tau(t)}_{\substack{\text{Total}\\\text{Applied Torque}}}$$

We want to look at the energy distribution of the system.

- **Multiply the Equation by Angular Velocity Term ω (方程两边乘以角速度)**

$$J\ddot{\theta}\cdot\dot{\theta} + B\dot{\theta}\cdot\dot{\theta} + K\theta\cdot\dot{\theta} = \tau(t)\cdot\omega$$

- **Integrate the above Equation w.r.t. Time (方程两边关于时间积分):**

$$\underbrace{\int_{t_0}^{t_1} J\ddot{\theta}\cdot\dot{\theta}\,dt}_{\Delta KE} + \underbrace{\int_{t_0}^{t_1} B\omega\cdot\omega\,dt}_{\int_{t_0}^{t_1} B\omega^2 dt \,\geq\, 0} + \underbrace{\int_{t_0}^{t_1} K\theta\cdot\dot{\theta}\,dt}_{\Delta PE} = \underbrace{\int_{t_0}^{t_1} \tau(t)\cdot\omega\,dt}_{W}$$

$\Downarrow$ **Change of kinetic energy (动能变化)**

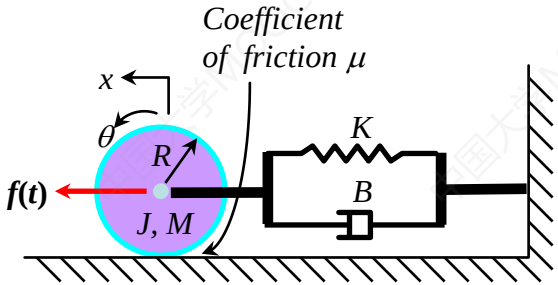
$\Downarrow$ **Energy dissipated by damper (通过阻尼器的能量耗散)**

$\Downarrow$ **Change of potential energy (势能变化)**

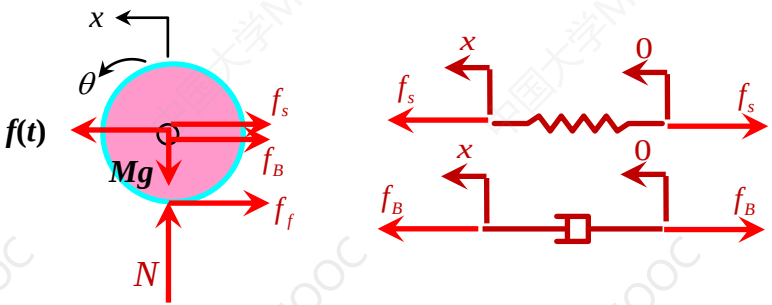
$\Downarrow$ **Total work done by the applied torque τ(t) from time t_0 to t_1 （作用力矩的总功)**

Example

- **Rolling without slipping** (不打滑滚动)



***FBD*:**



Elemental Laws:

$$M\ddot{x} = f - f_s - f_B - f_f \quad (1)$$

$$0 = N - Mg \quad (2)$$

$$J\ddot{\theta} = f_f R \quad (3)$$

$$f_s = K(x - 0) \quad (4)$$

$$f_B = B(\dot{x} - 0) \quad (5)$$

$$UV: x,\ f_s,\ f_B,\ f_f,\ N,\ \theta$$

$$\rightarrow x = R\theta \quad (6)$$

Example (cont.)

- **I/O Model** (Input: f, Output: x):

$$M\ddot{x} = f - f_s - f_B - f_f \quad (1)$$

$$0 = N - Mg \quad (2)$$

$$J\ddot{\theta} = f_f R \quad (3)$$

$$f_s = K(x - 0) \quad (4)$$

$$f_B = B(\dot{x} - 0) \quad (5)$$

$$x = R\theta \quad (6)$$

$$\Downarrow$$

$$(M + \frac{1}{R^2}J)\ddot{x} + B\dot{x} + Kx = f$$

Q: How would you decide whether or not the disk will slip? **(如何确定轮子打滑或不打滑)**

If

$$|f_f| = \left|\frac{J}{R}\ddot{\theta}\right| = \left|\frac{J}{R^2}\ddot{x}\right| < \mu_s N = \mu_s Mg$$

Then, no slip. **(不打滑)**

Q: How will the model be different if the disk rolls and slips? **(如果轮子既滚动又打滑，模型有何不同)**

When slips **(打滑时)** $x \neq R\theta$

But when the friction is known to be the moving friction **(如果摩擦力是已知的滑动摩擦力)**

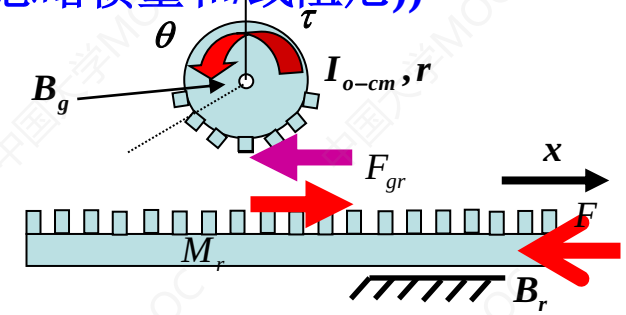
$$f_f = \mu_m N = \mu_m \mathrm{Mg} \quad (6)$$

So still 6 equations and 6 Uvs **(6个方程6个未知量)**

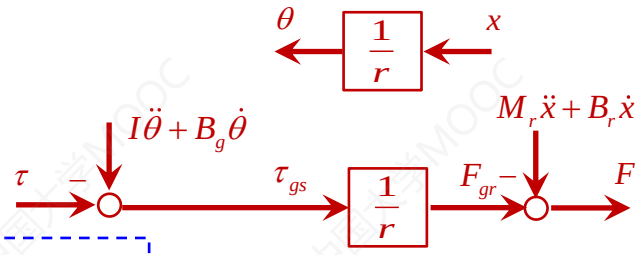
Example (coupled translation-rotation)

- **Real world (non-negligible inertias and/or damping)**

(实际(不忽略惯量和/或阻尼))



$$\begin{cases} I\ddot{\theta} = \tau - B_g\dot{\theta} - F_{gr}r \\ M_r\ddot{x} = F_{gr} - F - B_r\dot{x} \\ x = r\theta \end{cases}$$



$$\Rightarrow \quad \boxed{\underbrace{\left(M_r + \frac{I}{r^2}\right)}_{M_{eff}}\ddot{x} + \underbrace{\left(B_r + \frac{B_g}{r^2}\right)}_{B_{eff}}\dot{x} = \frac{\tau}{r} - F}$$

Think about performance of lever and gear train in real life

(考虑实际杆杠和齿轮的性能)

Summary

- Variables (变量)
- Basic Modeling Elements (基本建模元件)
- Interconnection Laws (互联关系)
- Derive Equation of Motion (EOM) (运动方程)